

DESIGNING FM RECEIVER

Using GNU Radio And RTL-SDR Dongle

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This article covers the designing of an FM receiver using GNU Radio software and RTL-SDR hardware on Ubuntu operating system (OS). With advancements in software-defined radio (SDR), more supports are available for SDR hardware to be used with computers. GNU Radio is a free software and provides signal processing capabilities. You can design the FM receiver using SDR with added features and functionalities, which are not available in basic radio receivers.

To design the FM receiver, you need RTL-SDR dongle (Fig. 1) with RTL chip, computer and GNU Radio live disk. GNU Radio Companion (GRC) is a block-based software design platform that makes it easy to work with radio systems by using only a block-like interface.

RTL-SDR dongle provides raw data to GNU Radio. All signal processing is performed using reconfigurable software. That is, GNU Radio is used for implementing the receiver by selecting signal processing blocks, which

process the incoming signal from RTL-SDR dongle to demodulate FM signal and to

get audio output from the computer speakers.

GNU Radio live disk

GNU Radio live disk provides options for quick and easy testing and experimentation with GNU Radio. The best part is that you do not need to change the settings in the system or the main OS. Drivers and support packages for RTL-SDR are preinstalled in GNU Radio disk, so it is convenient to work with the USB dongle with GNU Radio Companion.

A list of pre-installed SDR hardware drivers in GNU Radio live disk are given below.

1. Ettus Research UHD
2. Great Scott Gadgets HackRF products
3. Nuand BladeRF products
4. osmoSDR RTL-SDR driver
5. AirSpy

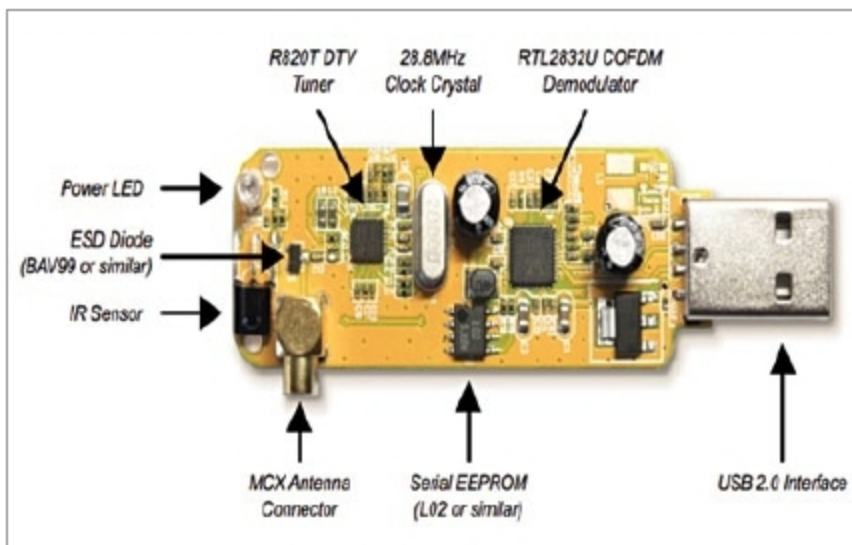


Fig. 1: RTL-SDR dongle



Fig. 2: GNU Radio Website QR Code

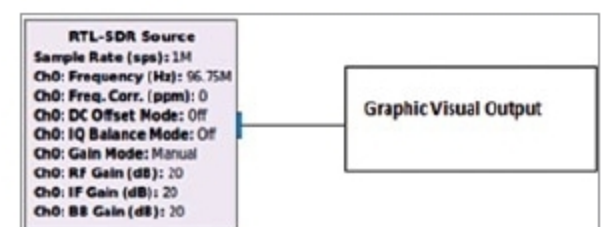


Fig. 4: Testing RTL-SDR support in GNU Radio



Fig. 3: GNU Radio live disk desktop and icon



Fig. 5: Antenna used in testing radio receiver